

SQL PORTFOLIO PROJECT 1

TechStore Sales Analysis Using SQL

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Language: SQL

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1. Introduction

This project demonstrates the use of SQL to analyze sales data for a fictional retail company called TechStore. The objective of the project was to answer business questions using SQL and provide insights that could assist management in making informed business decisions.

The project involved designing a relational database, creating tables, inserting sample data, writing SQL queries, and interpreting the results. Throughout the analysis, SQL techniques such as filtering, sorting, aggregate functions, grouping, joins, views, and subqueries were applied to solve real business problems.

This project was completed as part of my Data Analytics portfolio to demonstrate practical SQL skills that are relevant to a Junior Data Analyst role.

2. Business Scenario

TechStore is an electronics retail company that sells products such as laptops, keyboards, monitors, headphones and other technology products. The Sales Manager wants to better understand the company's sales performance and has requested an analysis of customer spending, product performance, and sales across different cities.

As a Junior Data Analyst, my responsibility was to extract meaningful information from the company's database using SQL and present insights that could support business decision-making.

3. Project Objectives

The objectives of this project were:

- Design a relational database using SQL.
- Create tables with primary and foreign key relationships.
- Populate the database with sample business data.
- Retrieve and filter information using SQL queries.
- Perform data aggregation and calculations.
- Analyze customer spending behavior.
- Identify top-performing products.
- Analyze sales performance across different cities.
- Create reusable SQL Views.
- Produce business recommendations based on data analysis.

4. Database Design

The database consists of four related tables.

Customers- Stores customer information.

Primary Key- CustomerID

Products- Stores product information including product prices and categories.

Primary Key- ProductID

Orders- Stores customer purchase transactions.

Primary Key- OrderID

Foreign Keys:

CustomerID → Customers(CustomerID)

ProductID → Products(ProductID)

Employees- Stores employee information used for practicing SQL queries involving filtering, sorting, grouping and aggregate functions.

5. Database Schema

The TechStore database was designed using a relational database model consisting of four interconnected tables.

Table1: Customers

Column Name	Data Type	Description
CustomerID	INT	Unique identifier for each customer (Primary Key)
CustomerName	VARCHAR(100)	Full name of the customer
City	VARCHAR(50)	Customer's city

Table2: Products

Column Name	Data Type	Description
ProductID	INT	Unique identifier for each product (Primary Key)
ProductName	VARCHAR(100)	Product name
Category	VARCHAR(50)	Product category
Price	DECIMAL(10,2)	Selling price of the product

Table3: Orders

Column Name	Data Type	Description
OrderID	INT	Unique identifier for each order (Primary Key)
CustomerID	INT	Customer placing the order (Foreign Key)
ProductID	INT	Product purchased (Foreign Key)
Quantity	INT	Number of units purchased
OrderDate	DATE	Date the order was placed

Table4: Employees

Column Name	Data Type	Description
EmployeeID	INT	Unique employee identifier (Primary Key)
FirstName	VARCHAR(50)	Employee first name
LastName	VARCHAR(50)	Employee surname
Department	VARCHAR(50)	Department
Salary	DECIMAL(10,2)	Employee salary

Entity Relationships

The relationship between the tables is illustrated below:

The Orders table acts as the central transaction table linking customers to the products they purchase.

6. SQL Skills Demonstrated

Throughout this project, several SQL concepts were applied to solve business problems.

Database Management

- Creating databases
- Creating tables
- Defining Primary Keys
- Defining Foreign Keys

Data Retrieval

- SELECT
- WHERE
- DISTINCT
- ORDER BY
- LIMIT

Aggregate Functions

- COUNT()
- SUM()
- AVG()

- MAX()
- MIN()

Data Grouping

- GROUP BY
- HAVING

Table Relationships

- INNER JOIN
- LEFT JOIN

SQL Objects

- CREATE VIEW
- Querying Views

Advanced SQL

- Business calculations
- Revenue calculations
- Average order value
- Basic subqueries

7. Business Questions

The Sales Manager requested answers to the following business questions.

Customer Analysis

1. Display all customers.

2. Count the total number of customers.
3. Display the top three highest-spending customers.
4. Identify customers who spent more than R10,000.
5. Calculate the average order value for each customer.

Product Analysis

6. Display products costing more than R1,000.
7. Identify the top five most expensive products.
8. Determine which product generated the highest revenue.
9. Determine which product sold the highest number of units.
10. Calculate the average product price by category.

Sales Analysis

11. Determine which city generated the highest revenue.
12. Calculate revenue generated by each product.
13. Count the number of orders placed by each customer.

SQL Objects

14. Create a reusable SQL View showing customer spending.
15. Use the SQL View to identify customers who spent more than R10,000.

8. SQL Solutions

Rather than listing every SQL query in this report, all SQL scripts used to complete this project are included in a separate file titled:

TechStore_SQL_Queries.sql

This script contains all SQL statements used throughout the project, including:

- Database creation
- Table creation
- Sample data insertion
- Data retrieval queries
- Aggregate calculations
- JOIN operations
- GROUP BY and HAVING queries
- View creation
- Business analysis queries

9. Business Insights

The SQL analysis produced several key findings that provide valuable insights into TechStore's sales performance.

Insight 1: High-Value Customers

The analysis identified Lerato Lehoko and Sarah Mokoena as the company's highest-spending customers. These customers contributed the highest total sales value and represent an important customer segment. Retaining high-value customers is essential because repeat customers often contribute significantly to overall revenue.

Insight 2: Top Revenue-Generating Products

The analysis showed that Laptops and Keyboards generated the highest revenue. These products play an important role in the company's overall sales performance and should be considered key products within the business.

Insight 3: Regional Sales Performance

Sales revenue differed across cities. The analysis showed that Pretoria generated the highest revenue, followed by Johannesburg. These findings indicate that customer demand varies by location and that regional sales performance should be considered when planning marketing and sales strategies.

Insight 4: Customer Purchasing Behaviour

By calculating customer spending and average order values, it became possible to identify customers who regularly make larger purchases. Understanding customer purchasing behavior allows businesses to create targeted marketing campaigns and customer loyalty programmes.

Insight 5: Value of SQL in Business Decision-Making

This project demonstrates how SQL can transform raw transactional data into meaningful business information. Using SQL queries, it was possible to answer important business questions regarding customers, products, revenue and regional performance without manually analyzing thousands of records.

10. Business Recommendations

Based on the analysis, the following recommendations are proposed.

Recommendation 1: Focus Marketing on High-Revenue Cities

Since Pretoria and Johannesburg generated the highest revenue, marketing campaigns should prioritize these locations. Increasing promotional activities in these cities may lead to higher sales and improved business growth.

Recommendation 2: Promote High-Performing Products

Laptops and Keyboards generated the highest revenue. The company should continue investing in these products through promotions, advertising and inventory management to maximize future sales.

Recommendation 3: Reward High-Value Customers

Customers who consistently spend the most money should be rewarded through loyalty programmes, exclusive discounts or personalized offers. Retaining existing high-value customers is generally more cost-effective than acquiring new ones.

Recommendation 4: Collect and Analyze Customer Feedback

Customer reviews and feedback should be collected regularly to better understand customer preferences. This information can be used to improve product quality, identify popular products and tailor marketing strategies for different customer groups and regions.

11. Skills Demonstrated

Throughout this project, I demonstrated the following technical and analytical skills:

Technical Skills

- MySQL Database Design
- Relational Database Concepts
- Primary and Foreign Keys
- Data Retrieval using SQL
- Data Filtering
- Sorting and Ranking
- Aggregate Functions
- GROUP BY and HAVING
- SQL JOINS
- SQL Views
- Basic Subqueries
- Business Calculations
- Query Optimisation Principles

Analytical Skills

- Business Problem Solving
- Customer Behaviour Analysis
- Revenue Analysis
- Product Performance Analysis
- Regional Sales Analysis
- Data Interpretation
- Business Reporting
- Business Recommendations

Professional Skills

- Logical Thinking
- Attention to Detail
- Problem Solving
- Critical Thinking
- Data Storytelling
- Technical Documentation

12. Conclusion

This project demonstrates the practical application of SQL to solve real business problems using a relational database. The project involved designing a database, creating relationships between tables, writing SQL queries, analyzing sales data and presenting actionable business insights. Through this analysis, customer spending patterns, product performance and regional sales trends were successfully identified. The findings were used to develop practical business recommendations that could support strategic decision-making.

Completing this project strengthened my understanding of SQL and reinforced the importance of using data to answer business questions. It also provided valuable hands-on experience that reflects the responsibilities of a Junior Data Analyst. This project represents the first completed project in my Data Analytics portfolio and serves as evidence of my ability to apply SQL to real-world business scenarios.